

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab task:-06



Submitted to:

Mr. Meesam Raza

Submitted by:

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Roll No: SP21379

Subject: Operating System

Department: BSSE-4-Afternoon

Date: 28-04-2024

Task 01

```
zulqarnain@lenovo:~/Desktop$ who
zulqarnain tty2          2024-04-27 08:50 (tty2)
zulqarnain@lenovo:~/Desktop$ who | cut -d" " -f1
zulqarnain
zulqarnain@lenovo:~/Desktop$ who | sort | uniq -u
zulqarnain tty2          2024-04-27 08:50 (tty2)
zulqarnain@lenovo:~/Desktop$ last
zulqarna tty2            tty2            Sat Apr 27 08:50    still logged in
reboot  system boot      6.5.0-26-generic  Sat Apr 27 08:47    still running
zulqarna tty2            tty2            Mon Apr 22 10:29 - crash (4+22:18)
reboot  system boot      6.5.0-26-generic  Mon Apr 22 10:27    still running
zulqarna tty2            tty2            Thu Mar 28 09:38 - crash (25+00:48)
zulqarna tty2            tty2            Thu Mar 28 09:38 - 09:38 (00:00)
reboot  system boot      6.5.0-26-generic  Thu Mar 28 09:36    still running
zulqarna tty2            tty2            Sun Mar 10 21:39 - crash (17+11:57)
reboot  system boot      6.5.0-25-generic  Sun Mar 10 21:37    still running
zulqarna tty2            tty2            Sat Mar 9 14:01 - crash (1+07:35)
reboot  system boot      6.5.0-25-generic  Sat Mar 9 14:00    still running
zulqarna tty2            tty2            Sat Mar 9 13:43 - down (00:17)
reboot  system boot      6.5.0-25-generic  Sat Mar 9 13:42 - 14:00 (00:18)

wtmp begins Sat Mar 9 13:42:15 2024
zulqarnain@lenovo:~/Desktop$ last | tac
wtmp begins Sat Mar 9 13:42:15 2024

reboot  system boot      6.5.0-25-generic  Sat Mar 9 13:42 - 14:00 (00:18)
zulqarna tty2            tty2            Sat Mar 9 13:43 - down (00:17)
reboot  system boot      6.5.0-25-generic  Sat Mar 9 14:00    still running
zulqarna tty2            tty2            Sat Mar 9 14:01 - crash (1+07:35)
reboot  system boot      6.5.0-25-generic  Sun Mar 10 21:37    still running
zulqarna tty2            tty2            Sun Mar 10 21:39 - crash (17+11:57)
reboot  system boot      6.5.0-26-generic  Thu Mar 28 09:36    still running
zulqarna tty2            tty2            Thu Mar 28 09:38 - 09:38 (00:00)
zulqarna tty2            tty2            Thu Mar 28 09:38 - crash (25+00:48)
reboot  system boot      6.5.0-26-generic  Mon Apr 22 10:27    still running
zulqarna tty2            tty2            Mon Apr 22 10:29 - crash (4+22:18)
reboot  system boot      6.5.0-26-generic  Sat Apr 27 08:47    still running
zulqarna tty2            tty2            Sat Apr 27 08:50    still logged in
zulqarnain@lenovo:~/Desktop$
```

```
zulqarnain@lenovo:~/Desktop$ cat > spelling.sed
s/enviroment/environment/g
zulqarnain@lenovo:~/Desktop$ cat>text.txt
envitoment is our enviromment
zulqarnain@lenovo:~/Desktop$ sed -f spelling.sed text.txt > text_file.txt
zulqarnain@lenovo:~/Desktop$ cat text_file.txt
envitoment is our enviromment
zulqarnain@lenovo:~/Desktop$ sed -f spelling.sed text.txt > text_file.txt | nl
zulqarnain@lenovo:~/Desktop$ cat text_file.txt | nl
1  envitoment is our enviromment
zulqarnain@lenovo:~/Desktop$
```

Task 02

```
zulqarnain@lenovo:~$ touch g.txt
zulqarnain@lenovo:~$ echo "this is sixth line" >>g.txt
zulqarnain@lenovo:~$ tr 'ABCDEF' '012345' g.txt
tr: extra operand 'g.txt'
Try 'tr --help' for more information.
zulqarnain@lenovo:~$ cat g.txt
this is sixth line
zulqarnain@lenovo:~$ /bin/ls -f
tut1.cpp                                .config                                .bash_logout                        .vboxclient-seamless-tty2-control.pid
..                                       .vboxclient-vmsvga-session-tty2-control.pid .vboxclient-vmsvga-session-tty2-service.pid Music
sonnet-18.txt.old                       Templates                             Pictures                           .vboxclient-seamless-tty2-service.pid
poem.txt.old                            .local                               .gnupg                             .vboxclient-draganddrop-tty2-service.pid
.vboxclient-hostversion-tty2-control.pid Documents                           .vboxclient-clipboard-tty2-service.pid Videos
.vboxclient-clipboard-tty2-control.pid secret-of-the-univers                .ssh                               .
message                                .sudo_as_admin_successful           .cache                             Downloads
.bashrc                                snap                                .bash_history                      Public
.lessht                                g.txt                              .profile
a.out                                  .vboxclient-draganddrop-tty2-control.pid Desktop

zulqarnain@lenovo:~$ less -f /bin/ls
zulqarnain@lenovo:~$
```

```
zulqarnain@lenovo:~$ od -t x /bin/ls
00000000 042577 043114 000402 000001 000000 000000 000000 000000
00000020 000003 000076 000001 000000 065240 000000 000000 000000
00000040 000100 000000 000000 000000 012050 000002 000000 000000
00000060 000000 000000 000100 000070 000015 000100 000037 000036
00000100 000006 000000 000004 000000 000100 000000 000000 000000
00000120 000100 000000 000000 000000 000100 000000 000000 000000
00000140 001330 000000 000000 000000 001330 000000 000000 000000
00000160 000010 000000 000000 000000 000003 000000 000004 000000
00000200 001430 000000 000000 000000 001430 000000 000000 000000
00000220 001430 000000 000000 000000 000034 000000 000000 000000
00000240 000034 000000 000000 000000 000001 000000 000000 000000
00000260 000001 000000 000004 000000 000000 000000 000000 000000
00000300 000000 000000 000000 000000 000000 000000 000000 000000
00000320 032130 000000 000000 000000 032130 000000 000000 000000
00000340 010000 000000 000000 000000 000001 000000 000005 000000
00000360 040000 000000 000000 000000 040000 000000 000000 000000
00000400 040000 000000 000000 000000 030221 000001 000000 000000
00000420 030221 000001 000000 000000 010000 000000 000000 000000
00000440 000001 000000 000004 000000 100000 000001 000000 000000
00000460 100000 000001 000000 000000 100000 000001 000000 000000
00000500 072130 000000 000000 000000 072130 000000 000000 000000
00000520 010000 000000 000000 000000 000001 000000 000006 000000
00000540 177720 000001 000000 000000 007720 000002 000000 000000
00000560 007720 000002 000000 000000 011250 000000 000000 000000
00000600 022560 000000 000000 000000 010000 000000 000000 000000
00000620 000002 000000 000006 000000 005130 000002 000000 000000
00000640 015130 000002 000000 000000 015130 000002 000000 000000
00000660 001000 000000 000000 000000 001000 000000 000000 000000
00000700 000010 000000 000000 000000 000004 000000 000004 000000
00000720 001470 000000 000000 000000 001470 000000 000000 000000
00000740 001470 000000 000000 000000 000060 000000 000000 000000
00000760 000060 000000 000000 000000 000010 000000 000000 000000
00010000 000004 000000 000004 000000 001550 000000 000000 000000
00010020 001550 000000 000000 000000 001550 000000 000000 000000
00010040 000104 000000 000000 000000 000104 000000 000000 000000
```

```
zulqarnain@lenovo:~$ od -t x1 /bin/ls
00000000 7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00
00000020 03 00 3e 00 01 00 00 00 a0 6a 00 00 00 00 00
00000040 40 00 00 00 00 00 00 00 28 14 02 00 00 00 00
00000060 00 00 00 00 40 00 38 00 0d 00 40 00 1f 00 1e
00000080 06 00 00 00 04 00 00 00 40 00 00 00 00 00 00
000000a0 40 00 00 00 00 00 00 00 40 00 00 00 00 00 00
000000c0 d8 02 00 00 00 00 00 d8 02 00 00 00 00 00
000000e0 08 00 00 00 00 00 00 00 03 00 00 00 04 00 00
00000100 18 03 00 00 00 00 00 00 18 03 00 00 00 00 00
00000120 18 03 00 00 00 00 00 00 1c 00 00 00 00 00 00
00000140 1c 00 00 00 00 00 00 00 01 00 00 00 00 00 00
00000160 01 00 00 00 04 00 00 00 00 00 00 00 00 00 00
00000180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
000001a0 58 34 00 00 00 00 00 00 58 34 00 00 00 00 00
000001c0 00 10 00 00 00 00 00 00 01 00 00 00 05 00 00
000001e0 00 40 00 00 00 00 00 00 00 40 00 00 00 00 00
00000200 00 40 00 00 00 00 00 00 91 30 01 00 00 00 00
00000220 91 30 01 00 00 00 00 00 10 00 00 00 00 00 00
00000240 01 00 00 00 04 00 00 00 80 01 00 00 00 00 00
00000260 00 80 01 00 00 00 00 00 80 01 00 00 00 00 00
00000280 58 74 00 00 00 00 00 00 58 74 00 00 00 00 00
000002a0 00 10 00 00 00 00 00 00 01 00 00 00 06 00 00
000002c0 d0 ff 01 00 00 00 00 d0 0f 02 00 00 00 00 00
000002e0 d0 0f 02 00 00 00 00 a8 12 00 00 00 00 00 00
00000300 70 25 00 00 00 00 00 00 10 00 00 00 00 00 00
00000320 02 00 00 00 06 00 00 00 58 0a 02 00 00 00 00
00000340 58 1a 02 00 00 00 00 58 1a 02 00 00 00 00 00
00000360 00 02 00 00 00 00 00 00 02 00 00 00 00 00 00
00000380 08 00 00 00 00 00 00 00 04 00 00 00 04 00 00
000003a0 38 03 00 00 00 00 00 38 03 00 00 00 00 00 00
000003c0 38 03 00 00 00 00 00 30 00 00 00 00 00 00 00
000003e0 30 00 00 00 00 00 00 08 00 00 00 00 00 00 00
00000400 04 00 00 00 04 00 00 68 03 00 00 00 00 00 00
00000420 68 03 00 00 00 00 68 03 00 00 00 00 00 00 00
00000440 44 00 00 00 00 00 44 00 00 00 00 00 00 00 00
00000460 04 00 00 00 00 00 53 e5 74 64 04 00 00 00 00
00000480 38 03 00 00 00 00 38 03 00 00 00 00 00 00 00
000004a0 38 03 00 00 00 00 30 00 00 00 00 00 00 00 00
000004c0 30 00 00 00 00 00 08 00 00 00 00 00 00 00 00
000004e0 50 e5 74 64 04 00 00 cc cd 01 00 00 00 00 00
00000500 cc cd 01 00 00 00 00 cc cd 01 00 00 00 00 00
00000520 6c 05 00 00 00 00 6c 05 00 00 00 00 00 00 00
00000540 04 00 00 00 00 00 51 e5 74 64 06 00 00 00 00
```

Task 03

```
zulqarnain@lenovo:~$ mkdir chunkD
zulqarnain@lenovo:~$ cd chunkD
zulqarnain@lenovo:~/chunkD$
zulqarnain@lenovo:~/chunkD$
zulqarnain@lenovo:~/chunkD$ touch chunks
zulqarnain@lenovo:~/chunkD$ cd ..
zulqarnain@lenovo:~$ less -f /bin/ls >>chunkD/chunks
zulqarnain@lenovo:~$ cd chunkD
zulqarnain@lenovo:~/chunkD$ ls
chunks
zulqarnain@lenovo:~/chunkD$ split -b 1k chunks chunk
zulqarnain@lenovo:~/chunkD$ ls
chunkaa chunkan chunkba chunkbn chunkca chunkcn chunkda chunkdn chunkea chunken chunkfa
chunkab chunkao chunkbb chunkbo chunkcb chunkco chunkdb chunkdo chunkeb chunkeo chunkfb
chunkac chunkap chunkbc chunkbp chunkcc chunkcp chunkdc chunkdp chunkec chunkep chunkfc
chunkad chunkaq chunkbd chunkbq chunkcd chunkcq chunkdd chunkdq chunked chunkeq chunkfd
chunkae chunkar chunkbe chunkbr chunkce chunkcr chunkde chunkdr chunkee chunker chunkfe
chunkaf chunkas chunkbf chunkbs chunkcf chunkcs chunkdf chunkds chunkef chunkes chunks
chunkag chunkat chunkbg chunkbt chunkcg chunkct chunkdg chunkdt chunkeg chunket
chunkah chunkau chunkbh chunkbu chunkch chunkcu chunkdh chunkdu chunkeh chunkeu
chunkai chunkav chunkbi chunkbv chunkci chunkcv chunkdi chunkdv chunkei chunkev
chunkaj chunkaw chunkbj chunkbw chunkcj chunkcw chunkdj chunkdw chunkej chunkew
chunkak chunkax chunkbk chunkbx chunkck chunkcx chunkdk chunkdx chunkek chunkex
chunkal chunkay chunkbl chunkby chunkcl chunkcy chunkdl chunkdy chunkel chunkey
chunkam chunkaz chunkbm chunkbz chunkcm chunkcz chunkdm chunkdz chunkem chunkez
zulqarnain@lenovo:~/chunkD$
```